Two-Block Phase Diagram: Supplementary Experiments and Seed Dependence

Purpose. This supplement records the training-history and numerical-control experiments omitted from the concise theory comparison. The full sweep contains three independent initialization seeds (0, 1, 2), 72 decay cells per seed, and two matched protocols per cell.

The two protocols were both run. *Joint* training starts from the random initialization and trains both degree-2 parents and the degree-4 target simultaneously, with every parameter trainable. *Parent-warm* starts from the same seeded base model, first trains only E, V_1, W_{d2} on the two parent targets read from h_1, and then unfreezes the entire model and trains the identical final objective as joint. There are 216 runs of each protocol (432 total). Thus protocol differences isolate training history, not architecture or final loss.

History changes the circuit. The two protocols selected different phase labels in 128 of 216 matched (seed, $\lambda_0, \lambda_1/\lambda_2$) comparisons (59.3%). Of those 128 changes, 122 were $H_D \to H_C$ from joint to parent-warm, three were suppressed$\to H_C$, and three were suppressed$\to H_D$. Parent warming therefore exposes a broad compositional basin that joint training almost always misses. It is not universally the lower-objective basin: parent-warm had lower final objective in 90/216 comparisons (41.7%), although its mean objective difference was -0.0488.

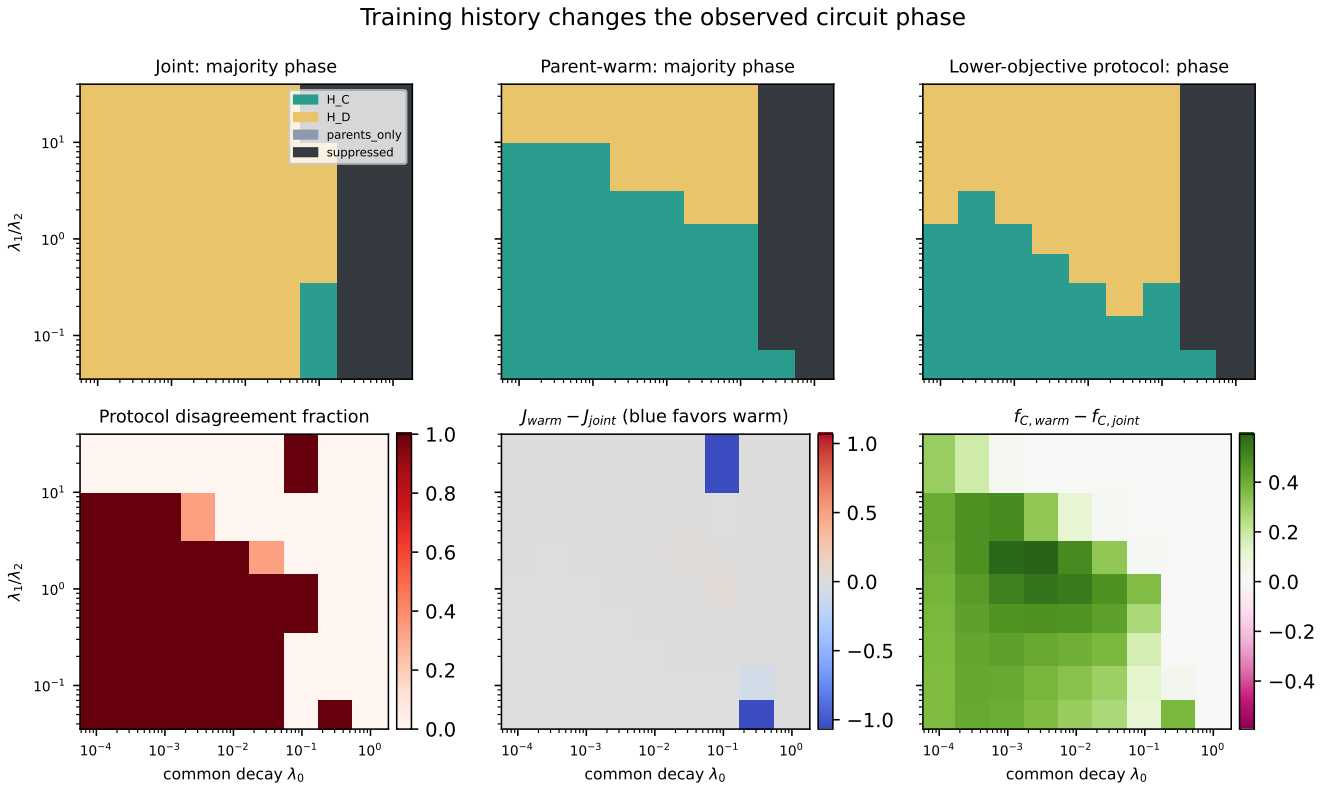


Figure 1: **Protocol-level phase diagrams.** Top: majority phase over three seeds for joint, parent-warm, and the lower-objective protocol used by the main report. Bottom: matched-seed disagreement, parent-warm minus joint objective, and the increase in total $2+2 \to 4$ share. The near-unity disagreement band is the compositional accessibility window opened by parent pretraining.

Degree-4-only objective comparison. To separate target fitting from the parent losses and shared hidden-weight costs, define

$$L_4 = \frac{\gamma_4}{2}\,\mathrm{MSE}_4, \qquad J_4^{\mathrm{spec}} = L_4 + \frac{\lambda_4}{2}\|w_4\|^2.$$

The first quantity is the degree-4 data term alone; the second adds the only regularizer uniquely attributable to the degree-4 target. Hidden-weight decay cannot be assigned unambiguously to one task and is therefore excluded. Here $\gamma_4 = 2$, so L_4 equals the reported degree-4 MSE.

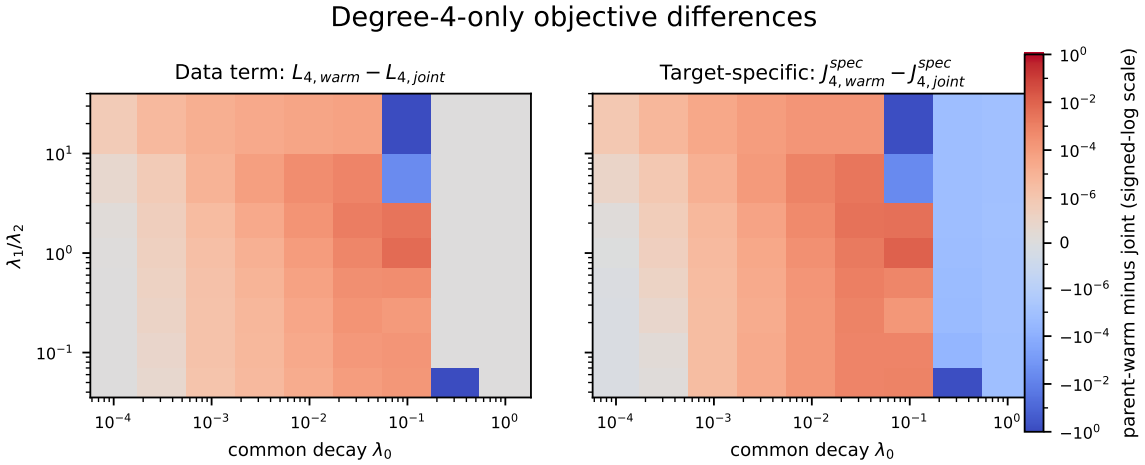


Figure 2: **Degree-4-only parent-warm minus joint differences, averaged over the three matched seeds.** Blue means parent-warm has the smaller value and red means joint does. The signed-log color normalization is linear inside $|\Delta| < 10^{-7}$ and logarithmic outside it, allowing differences from 10^{-9} to order one to remain visible. Across most active cells both protocols fit parity nearly perfectly; parent-warm usually has a slightly larger L_4 even when it forms the desired circuit. The isolated order-one blue cells arise when joint training suppresses the degree-4 task while parent warming keeps it active.

This comparison also explains why the lower-total-objective map cannot be inferred from degree-4 fit alone: its selection additionally includes the two parent losses and all five weight-decay terms. Circuit identity and the last few digits of degree-4 MSE are distinct diagnostics. Parent-warm has smaller L_4 in only 8/216 matched runs and smaller J_4^{spec} in 70/216. Nevertheless, the mean differences are negative because the few order-one joint-suppression failures outweigh the many positive differences of only 10^{-9}–10^{-3}.

Seed-by-seed robustness.

Seed	Joint: H_C/H_D/suppressed	Parent-warm: H_C/H_D/suppressed	Disagreement
0	3/52/17	46/11/15	44/72 (61.1%)
1	3/52/17	44/13/15	42/72 (58.3%)
2	3/52/17	44/13/15	42/72 (58.3%)
All	9/156/51	134/37/45	128/216 (59.3%)

The joint maps have identical phase counts for all three seeds. Parent-warm seeds 1 and 2 are identical in phase count, while seed 0 differs by only two cells. The shape of the compositional window is therefore reproducible; the large protocol effect is not caused by a single favorable seed. Seed variation is concentrated near the staircase boundary between H_C and H_D and near task suppression.

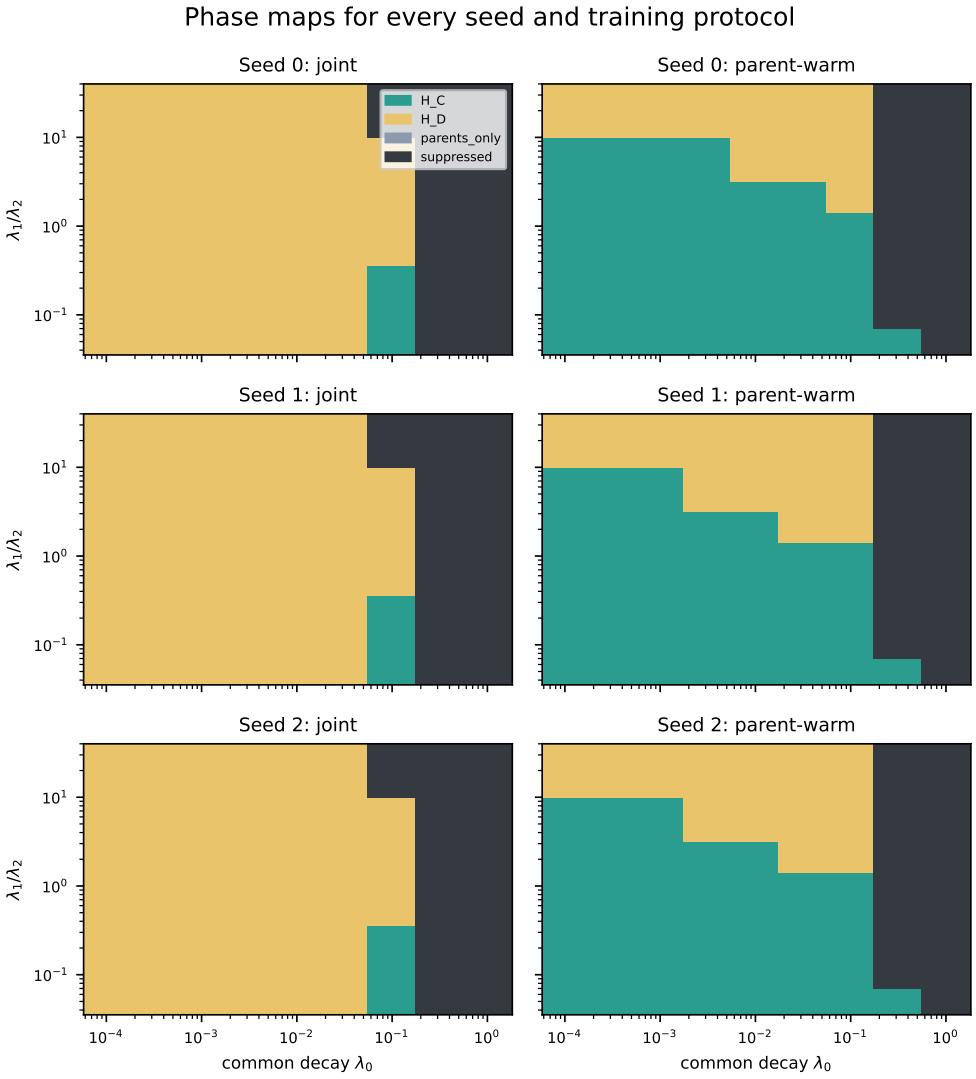


Figure 3: **All six empirical phase maps.** Each row fixes the random initialization; the left and right panels differ only by parent pretraining before the common fully unfrozen objective. The repeated left/right contrast makes the history dependence directly visible.

Numerical control 1: failed uniform-learning-rate launch. The first width-1024 launch used float64 full-batch SGD with momentum 0.9 and learning rate 0.01 for every parameter group. This is incompatible with the muP readout initialization $W = O(1/N)$: individual readout gradients are $O(1)$, so the first updates overwhelm their initial scale. Parent-pretraining states became non-finite and every worker raised a non-finite main objective at step 8. No valid phase cell was produced. This establishes a numerical requirement—scale the readout learning rate as $1/N$—not a circuit-phase result.

Numerical control 2: corrected width-1024 smoke test. Before restarting the sweep we tested seed 0 at $\lambda_0 = 0.01$ and $\lambda_1/\lambda_2 = 1$, using $\eta_{E,V} = 0.01$, $\eta_W = 0.01/1024$, 1,000 parent-warmup steps, and 1,000 final-objective steps.

Protocol	Phase	Objective	A_4	R	f_C	$\|\nabla J\|$
Joint	H_D	2.29699	0.9453	0.4454	0.1131	1.0490
Parent-warm	H_C	2.28398	0.9445	0.5879	0.5844	0.9698

Both runs stayed finite, and the parent-warm run already entered the compositional basin. Because the gradient norms were still order one, this was only a stability and basin test; it was not used as stationary phase evidence.

Finite-time qualification. Only 99/432 full-sweep runs met the plateau criterion; 333 reached the 50k-step cap. The phase maps therefore describe circuits found by the stated finite-time protocols, while the lower-objective aggregation is only a best-observed proxy for the zero-temperature theory.

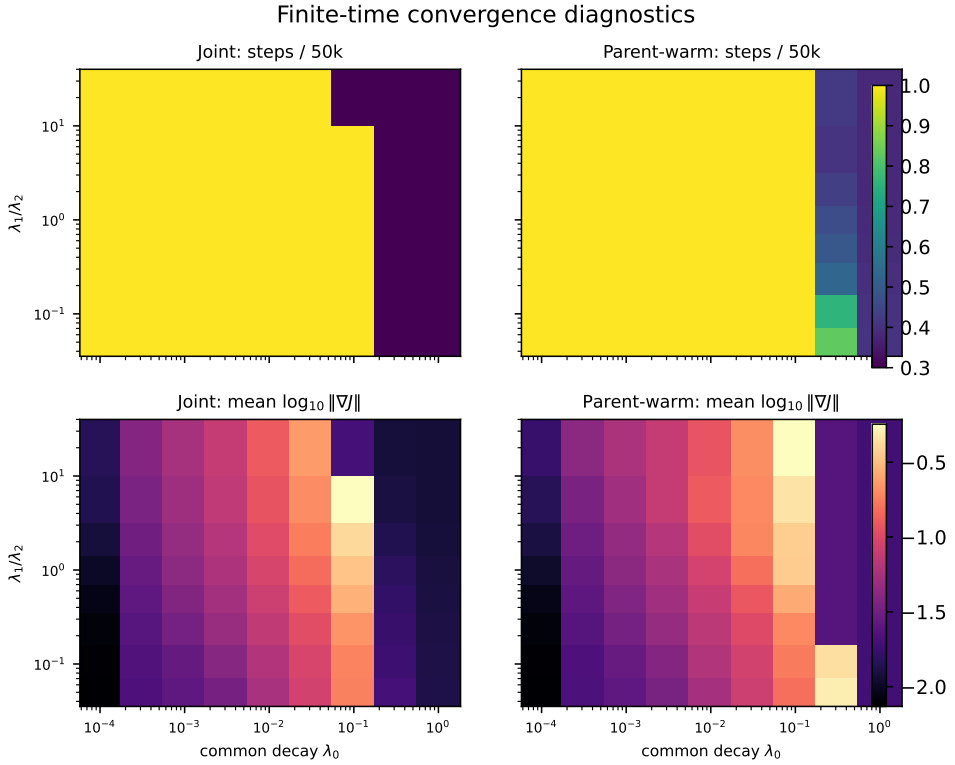


Figure 4: **Convergence diagnostics.** Mean training fraction and final raw gradient norm over the three seeds. Weak and moderate decay usually consume the full budget; rapid termination is concentrated in strongly regularized or suppressed cells.